ADVANCED EXCEL – LOGSHEET

Institute: Prakshal IT Academy, Gandhinagar

Module 1: Excel Fundamentals

1. Introduction to Microsoft Excel
2. Excel Interface and Workbook Management
3. Rows, Columns, Cells and Cell References
4. Data Entry and Editing
5. Saving and Managing Excel Files

Module 2: Excel Formatting

1. Font, Border and Alignment
2. Number and Date Formatting
3. Custom Number Formats
4. Conditional Formatting
5. Page Layout and Printing

Module 3: Basic Operations

1. Copy, Cut, Paste and Paste Special
2. AutoFill and Flash Fill
3. Find and Replace
4. Freeze Panes and Split Windows
5. Worksheet Management

Module 4: Formulas and References

1. Introduction to Excel Formulas
2. Relative and Absolute References
3. Mixed Cell References
4. Mathematical Operators
5. Formula Copying and Troubleshooting

Module 5: Basic Functions

1. SUM() and AVERAGE()
2. MIN() and MAX()
3. COUNT() and COUNTA()
4. ROUND() and ROUNDUP()

5. Practical Marksheet Preparation

Module 6: Logical Functions

1. IF() Function
2. Nested IF() and IFS()
3. AND() and OR()
4. IFERROR() and IFNA()
5. Pass/Fail and Grade Calculation

Module 7: Conditional Functions

1. SUMIF() and SUMIFS()
2. COUNTIF() and COUNTIFS()
3. AVERAGEIF() and AVERAGEIFS()
4. MAXIFS() and MINIFS()
5. Conditional Sales Analysis

Module 8: Text Functions

1. LEFT(), RIGHT() and MID()
2. LEN(), TRIM() and CLEAN()
3. UPPER(), LOWER() and PROPER()
4. CONCAT() and TEXTJOIN()
5. FIND(), SEARCH() and SUBSTITUTE()

Module 9: Date and Time

1. TODAY() and NOW()
2. DATE(), DAY(), MONTH() and YEAR()
3. DATEDIF() Function
4. WORKDAY() and NETWORKDAYS()
5. Attendance and Experience Calculation

Module 10: Lookup Functions

1. VLOOKUP() Function
2. HLOOKUP() Function
3. XLOOKUP() Function
4. INDEX() and MATCH()
5. Employee and Product Data Search

Module 11: Data Validation

1. Creating Drop-Down Lists
2. Number and Date Restrictions
3. Input Messages and Error Alerts
4. Dependent Drop-Down Lists
5. Automated Data Entry Form

Module 12: Data Cleaning

1. Removing Duplicate Records
2. Handling Blank Cells
3. Text to Columns
4. Flash Fill and Find & Replace
5. Correcting Dates and Numbers

Module 13: Data Sorting and Filtering

1. Basic and Multi-Level Sorting
2. AutoFilter
3. Text, Number and Date Filters
4. Advanced Filter
5. Practical Employee Data Analysis

Module 14: Excel Tables

1. Creating and Formatting Tables
2. Structured References
3. Table Sorting and Filtering
4. Total Row and Calculations
5. Inventory Management Sheet

Module 15: PivotTables

1. Creating PivotTables
2. PivotTable Fields and Layout
3. Grouping and Summarizing Data
4. Sorting and Filtering PivotTables
5. Monthly Sales Report

Module 16: Advanced PivotTables

1. Calculated Fields
2. Show Values As
3. Running Totals and Percentages
4. Slicers and Timelines
5. PivotCharts and Interactive Reports

Module 17: Charts and Visualization

1. Column and Bar Charts
2. Line and Pie Charts
3. Area and Scatter Charts
4. Combo Charts
5. Chart Formatting and Labels

Module 18: Advanced Charts

1. Dynamic Charts
2. Sparklines
3. Histogram and Pareto Charts
4. Secondary Axis
5. Interactive Chart Reports

Module 19: What-If Analysis

1. Goal Seek
2. Scenario Manager
3. One-Variable Data Table
4. Two-Variable Data Table
5. Profit and Break-Even Analysis

Module 20: Dashboard Development

1. Dashboard Planning and Layout
2. KPI Cards and Summary Metrics
3. PivotCharts and Slicers
4. Interactive Dashboard Filters
5. Sales Dashboard Project

Module 21: Power Query Basics

1. Introduction to Power Query

2. Importing Excel and CSV Files
3. Removing Errors and Duplicates
4. Splitting and Merging Columns
5. Loading Cleaned Data

Module 22: Advanced Power Query

1. Append Queries
2. Merge Queries
3. Combining Multiple Files
4. Group By and Custom Columns
5. Refreshing Automated Reports

Module 23: Excel Productivity and Protection

1. Named Ranges
2. Workbook and Worksheet Protection
3. Hyperlinks, Comments and Notes
4. Print Area and Page Breaks
5. Workbook Review and Presentation

Module 24: Macros and VBA

1. Introduction to Macros
2. Recording and Running Macros
3. Assigning Macros to Buttons
4. Introduction to VBA Editor
5. Basic VBA Automation

Module 25: Add-ins and AI

1. Managing Excel Add-ins
2. Analysis ToolPak
3. Descriptive Statistics
4. AI-Assisted Formula Writing
5. Microsoft Copilot in Excel (If Available)

Module 26: Business Reporting

1. Student Result Report
2. Employee Attendance Report

3. Payroll Management Report
4. Inventory Management Report
5. Monthly Sales and Expense Report

Module 27: Final Practical Project

1. Project Planning and Data Collection
2. Data Cleaning and Preparation
3. Formula-Based Calculations
4. PivotTables and Charts
5. Dashboard Creation and Submission

Module 28: Final Assessment

1. Formula Practical Test
2. Data Cleaning Practical Test
3. Lookup and PivotTable Test
4. Dashboard Development Test
5. Final Project Presentation